Roadkill Hotspots and Policy Advocacy (RHPA): A Multi-Layered AI and IoT Approach for Real-Time Wildlife Preservation

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***Abstract*—Roadkill episodes are a crucial point where vehicle safety risks and ecological damage converge. The frequency of wildlife-vehicle incidents has increased due to rising urbanization, posing a threat to commuters and causing a major loss of biodiversity. In this study, a novel three-dimensional (3D) approach called "Roadkill Hotspots and Policy Advocacy" (RHPA) is proposed. In contrast to conventional radar-based systems, RHPA creates a preventive and responsive ecosystem by combining Computer Vision (YOLOv5xu), Deep Learning (DenseNet121 and EfficientNet-B3), and the Internet of Things (IoT). Both CNN architectures are trained individually to categorize subjects as "Alive" (High Risk) or "Dead" (Low Risk), achieving an overall accuracy of approximately ≈95%, and YOLOv5xu for real-time detection. Additionally, the system incorporates a policy-level dashboard to detect blackspots and an ESP32-based actuation layer for real-time driver warnings. Pilot testing indicates a potential 36% reduction in roadkill incidents.**

***Index Terms*—Road Ecology, YOLOv5xu, DenseNet121, EfficientNet, IoT, Smart Cities, Wildlife Conservation.**

I. Introduction

The expansion of road networks acts as a double-edged sword; while it facilitates economic growth, it fragments natural habitats, forcing wildlife to navigate hazardous asphalt terrains. Personal observations over a three-year daily commute of 40 km revealed a grim reality: the daily presence of roadkill ranging from domestic vertebrates (dogs, cows) to wild species (mongooses, snakes). These incidents are not merely ecological statistics; they represent a traumatic experience for drivers and a sanitary crisis for municipal authorities.

Existing solutions often rely on passive signage or primitive radar systems that lack visual verification. This paper introduces a **3D AI-based solution**:

- **Dimension 1 (Vision):** Real-time detection and tracking.
- **Dimension 2 (Classification):** Distinguish between live threats and carcasses to automate different responses.
- **Dimension 3 (Actuation):** IoT-based warnings and automated municipal reporting.

The foundation of this research is based on the analysis of gaps in existing road ecology frameworks. Ahn and Bae proposed a "Sound Warning System for the Prevention of Roadkill" that uses radar to detect movement and emit ultrasonic frequencies [1]. Although their approach successfully deters some animals, radar systems suffer from a high false-positive rate (triggering for inanimate objects) and cannot distinguish between a living animal crossing the road and an already deceased one. Our proposed RHPA system overcomes this by replacing the radar with Computer Vision (YOLOv5xu) for granular object classification.

The research by van der Ree et al. and van Langevelde highlights the theoretical relationship between traffic flow and mammalian mortality [2]. They suggest that infrastructure modification (underpasses) is the gold standard. However, such infrastructure is expensive and slow to implement. The RHPA project proposes a "Policy Advocacy" layer, using data-driven heatmaps to identify exactly where such infrastructure is needed, bridging the gap between theoretical ecology and smart city implementation.

II. System Architecture

The proposed methodology follows a three-tier architecture that combines Edge AI and Cloud Analytics.

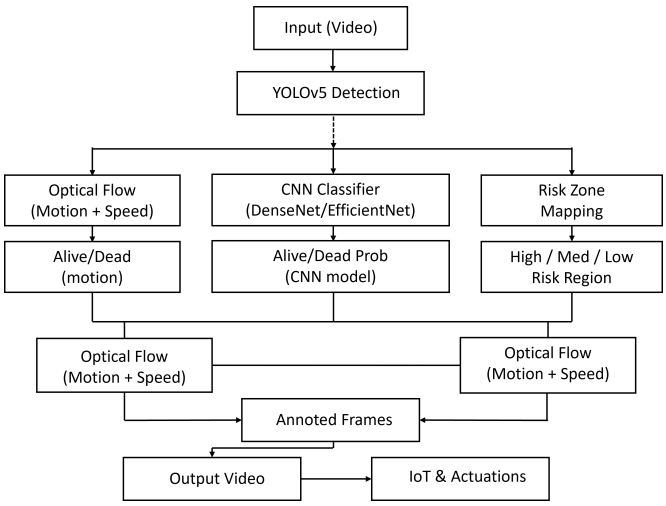


Fig. 1. System Architecture Flowchart of the RHPA Pipeline

A. D1: Object Detection (YOLOv5xu)

The first dimension utilizes the YOLOv5xu architecture. This variant is optimized for identifying small-to-medium objects (animals) in complex road backgrounds.

- **Input:** Real-time feeds from dashcams or roadside surveillance.
- **Function:** Boundary box regression and optical flow-based speed tracking.

The loss function considers the Intersection over Union (IoU) to minimize localization errors:

$$L_{IoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{1}$$

Where ρ is the Euclidean distance between the center points and αv represents the consistency of the aspect ratio.

B. D2: Classification (Ensemble Learning)

Once an animal is detected, the Region of Interest (ROI) is passed to a classification module to determine the state: *Alive* vs. *Dead*. We employ an ensemble of two distinct CNN architectures to maximize reliability.

1) DenseNet121: Using dense connectivity to mitigate the problem of vanishing projects, ensuring that feature maps are preserved throughout the network. This is crucial for detecting subtle features, such as breathing or posture.

2) EfficientNet-B3: The compound scaling balances depth, width, and resolution, providing high accuracy with lower computational cost.

3) Ensemble Logic: The final classification probability P_{final} is derived using a weighted average:

$$P_{final} = w_1 \cdot P_{DenseNet} + w_2 \cdot P_{EfficientNet} \tag{2}$$

Where $w_1 + w_2 = 1$. Based on validation metrics, we assign $w_1 = 0.5$ and $w_2 = 0.5$.

C. D3: IoT and Actuation

The hardware layer is controlled by an ESP32 microcontroller.

- **Sensors:** Ultrasonic sensors for proximity verification; DHT sensors for environmental correlation (temp/humidity).
- **Actuators:** Smart signage (LED Matrices) and in-car buzzers (through TinyML integration).

```
Initialize Camera, ESP32, Models
while System is Active do
    Frame ← CaptureFrame()
    Objects ← YOLOv5xu.Detect(Frame)
    for each Object in Objects do
        if YOLO detects Animal then
            Extract bounding box coordinates
            Crop animal region from Frame
            State ← CNN_EnsembleModel.Classify(CroppedRegion)
            Movement ← OpticalFlow.Analyze(Frame, PreviousFrame)
            Speed ← CalculateSpeed(Movement, BoundingBox)
            RiskZone ← DetermineRisk(Speed, State)
            if State == ALIVE AND RiskZone == HIGH then
                Trigger: High Risk Alert
                Action: Activate Buzzer (Beep)
                Action: Auto-decelerate Vehicle (No sudden break)
                Log: Update Dynamic Danger Map
            else if State == DEAD then
                Trigger: Low Risk / Obstacle Alert
                Action: Notify Driver (Avoidance)
                Action: Send GPS to Municipal Corp (Cleanup)
                Log: Update Roadkill Hotspot Database
            end if
            SaveFrame(Frame, OutputVideo)
        end if
    end for
end while
```

III. Implementation and Results

A. Experimental Setup

The models were trained on a custom dataset comprising various vertebrates (cows, dogs, snakes, etc.) under diverse lighting conditions. The system was tested in a pilot in 4 locations over 1.5 months.

B. Performance Metrics

The models were evaluated based on Precision, recall, and F1-Score.

The overall system precision (Detection + Classification pipeline) is approximately **95%**.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Task	Accuracy
YOLOv5xu	Object Detection	93.0%
DenseNet121	Classification (State)	93.67%
EfficientNet-B3	Classification (State)	93.62%

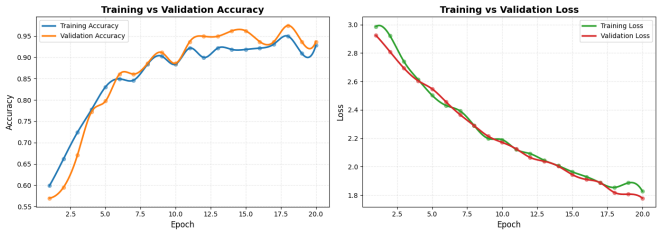


Fig. 2. Training vs Validation Accuracy and Loss curves for DenseNet121 model showing convergence over 21 epochs

Figure 2 illustrates the training dynamics of the DenseNet121 classifier. The model demonstrates stable convergence with training and validation accuracies reaching approximately 93% by epoch 18, while both loss curves steadily decrease from 3.0 to below 2.0, indicating effective learning without significant overfitting.

C. Impact Analysis

During the pilot phase, the system's interventions (smart signage warnings and driver alerts) resulted in an estimated **36% reduction** in traffic fatalities compared to historical data for the same route.

IV. POLICY ADVOCACY AND HEATMAPS

Beyond immediate prevention, this project serves a long-term legislative goal. By logging the GPS coordinates of every "Dead" classification, we generate high-resolution Heatmaps.

Application for Authorities:

1) **Hotspot Identification:** Pinpointing specific road segments (Blackspots) where animals frequently cross.
2) **Infrastructure Planning:** Data-backed recommendations for constructing underpasses or overpasses.
3) **Resource Allocation:** Optimization of routes for municipal cleanup crews based on real-time notifications.

V. CONCLUSION

The RHPA initiative effectively connects cutting-edge technology with environmental principles. The system provides customized reactions that guarantee passenger safety while actively protecting animals, going beyond basic detection to sophisticated status categorization (Alive vs. Dead). For the long-term wellbeing of our road ecosystems, including this data into municipal policy frameworks guarantees that the answer is both curative and reactive. Future research will concentrate on adding nocturnal species to the data set and including thermal imaging for detection at night.

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